

Problem 1

Show the following:

- (a) $\mathbb{E}(aY + bZ \mid X) = a\mathbb{E}(Y \mid X) + b\mathbb{E}(Z \mid X)$ for $a, b \in \mathbb{R}$,
- (b) $\mathbb{E}(Y \mid X) \geq 0$ if $Y \geq 0$,
- (c) $\mathbb{E}(1 \mid X) = 1$,
- (d) if X and Y are independent then $\mathbb{E}(Y \mid X) = \mathbb{E}(Y)$,
- (e) (‘pull-through property’) $\mathbb{E}(Yg(X) \mid X) = g(X)\mathbb{E}(Y \mid X)$ for any suitable function g ,
- (f) (‘tower property’) $\mathbb{E}\{\mathbb{E}(Y \mid X, Z) \mid X\} = \mathbb{E}(Y \mid X) = \mathbb{E}\{\mathbb{E}(Y \mid X) \mid X, Z\}$.

(a) Linearity: By definition, for discrete random variables,

$$\mathbb{E}(aY + bZ \mid X = x) = \sum_{y,z} [ay + bz] \mathbb{P}(Y = y, Z = z \mid X = x).$$

By the distributive property of summation, we have:

$$\mathbb{E}(aY + bZ \mid X = x) = a \sum_{y,z} y \mathbb{P}(Y = y, Z = z \mid X = x) + b \sum_{y,z} z \mathbb{P}(Y = y, Z = z \mid X = x).$$

Notice that for fixed y ,

$$\sum_z \mathbb{P}(Y = y, Z = z \mid X = x) = \mathbb{P}(Y = y \mid X = x),$$

and similarly,

$$\sum_y \mathbb{P}(Y = y, Z = z \mid X = x) = \mathbb{P}(Z = z \mid X = x).$$

Thus, we obtain:

$$\begin{aligned} \mathbb{E}(aY + bZ \mid X = x) &= a \sum_y y \mathbb{P}(Y = y \mid X = x) + b \sum_z z \mathbb{P}(Z = z \mid X = x) \\ &= a \mathbb{E}(Y \mid X = x) + b \mathbb{E}(Z \mid X = x). \end{aligned}$$

Since this holds for every x , we have proven that

$$\mathbb{E}(aY + bZ \mid X) = a \mathbb{E}(Y \mid X) + b \mathbb{E}(Z \mid X).$$

(b) Nonnegativity:

If $Y \geq 0$, then for any fixed x each term in

$$\mathbb{E}(Y \mid X = x) = \sum_y y \mathbb{P}(Y = y \mid X = x)$$

is nonnegative, so

$$\mathbb{E}(Y | X) \geq 0.$$

(c) Expectation of a Constant:

Setting $Y \equiv 1$,

$$\mathbb{E}(1 | X = x) = \sum_{y,z} 1 \cdot \mathbb{P}(Y = y, Z = z | X = x) = \sum_{y,z} \mathbb{P}(Y = y, Z = z | X = x) = 1,$$

since the sum of the conditional probabilities equals 1. Thus,

$$\mathbb{E}(1 | X) = 1.$$

(d) Independence:

If X and Y are independent, then for all x and y

$$\mathbb{P}(Y = y | X = x) = \mathbb{P}(Y = y).$$

Thus,

$$\mathbb{E}(Y | X = x) = \sum_y y \mathbb{P}(Y = y | X = x) = \sum_y y \mathbb{P}(Y = y) = \mathbb{E}(Y).$$

Hence,

$$\mathbb{E}(Y | X) = \mathbb{E}(Y).$$

(e) Pull-Through Property:

Let g be any function such that $g(X)$ is $\sigma(X)$ -measurable. Then for any fixed x ,

$$\begin{aligned} \mathbb{E}(Y g(X) | X = x) &= \sum_{y,z} y g(x) \mathbb{P}(Y = y, Z = z | X = x) \\ &= g(x) \sum_{y,z} y \mathbb{P}(Y = y, Z = z | X = x) \\ &= g(x) \mathbb{E}(Y | X = x). \end{aligned}$$

Thus,

$$\mathbb{E}(Y g(X) | X) = g(X) \mathbb{E}(Y | X).$$

(f) Tower Property:

$$\mathbb{E}\{\mathbb{E}(Y|X, Z)|X = x\} = \sum_z \left\{ \sum_y y \mathbb{P}(Y = y|X = x, Z = z) \mathbb{P}(X = x, Z = z|X = x) \right\}$$

$$\begin{aligned}
 &= \sum_z \sum_y y \frac{\mathbb{P}(Y = y, X = x, Z = z)}{\mathbb{P}(X = x, Z = z)} \cdot \frac{\mathbb{P}(X = x, Z = z)}{\mathbb{P}(X = x)} \\
 &= \sum_y y \mathbb{P}(Y = y | X = x) \\
 &= \mathbb{E}\{\mathbb{E}(Y | X) | X = x, Z = z\}
 \end{aligned}$$

Problem 2

Conditional variance formula.

How should we define $\text{var}(Y | X)$, the conditional variance of Y given X ?

Show that $\text{var}(Y) = \mathbb{E}(\text{var}(Y | X)) + \text{var}(\mathbb{E}(Y | X))$.

Problem 3

- (1) Let X and Y be independent discrete random variables, and let $g, h : \mathbb{R} \rightarrow \mathbb{R}$. Show that $g(X)$ and $h(Y)$ are independent.
- (2) Show that two discrete random variables X and Y are independent if and only if $f_{X,Y}(x, y) = f_X(x)f_Y(y)$ for all $x, y \in \mathbb{R}$.
- (3) More generally, show that X and Y are independent if and only if $f_{X,Y}(x, y)$ can be factorized as the product $g(x)h(y)$ of a function of x alone and a function of y alone.

Problem 4

Stein's identity. Let X be $N(\mu, \sigma^2)$.

Show that $\mathbb{E}\{(X - \mu)g(X)\} = \sigma^2 \mathbb{E}(g'(X))$ when both sides exist.

Problem 5

- (a) Show that $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$, and deduce that

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-\frac{(x - \mu)^2}{2\sigma^2}\right\}, \quad -\infty < x < \infty,$$

is a density function if $\sigma > 0$.

- (b) Calculate the mean and variance of a standard normal variable.
- (c) Show that the $N(0, 1)$ distribution function Φ satisfies

$$(x^{-1} - x^{-3})e^{-\frac{1}{2}x^2} < \sqrt{2\pi}[1 - \Phi(x)] < x^{-1}e^{-\frac{1}{2}x^2}, \quad x > 0.$$

These bounds are of interest because Φ has no closed form.

- (d) Let X be $N(0, 1)$, and $a > 0$. Show that $\mathbb{P}(X > x + a/x \mid X > x) \rightarrow e^{-a}$ as $x \rightarrow 0$.

Problem 6

- (a) Let X have a continuous distribution function F . Show that
- (i) $F(X)$ is uniformly distributed on $[0, 1]$,
 - (ii) $-\log F(X)$ is exponentially distributed.
- (b) A straight line l touches a circle with unit diameter at the point P which is diametrically opposed on the circle to another point Q . A straight line QR joins Q to some point R on l . If the angle $\widehat{PQR}$ between the lines PQ and QR is a random variable with the uniform distribution on $[-\frac{1}{2}\pi, \frac{1}{2}\pi]$, show that the length of PR has the Cauchy distribution (this length is measured positive or negative depending upon which side of P the point R lies).
- (c) Let the net scores in the two halves of a game between teams A and B be independent and identically distributed random variables X, Y that are symmetric about 0 and have a continuous distribution function F . Team A wins (respectively, loses) if $X + Y > 0$ (respectively, $X + Y \leq 0$). Find the probability that A wins conditional on the half-time score X .

Problem 7

Secretary/marriage problem. You are permitted to inspect the n prizes at a fête in a given order, at each stage either rejecting or accepting the prize under consideration. There is no recall, in the sense that no rejected prize may be accepted later. It may be assumed that, given complete information, the prizes may be ranked in a strict order of preference, and that the order of presentation is independent of this ranking. Find the strategy which maximizes the probability of accepting the best prize, and describe its behaviour when n is large.